

Say, Echo, Do:

Strategic Speech, Media Echo, and the Economics of False Alarms

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Abstract. Markets hear three voices: what institutions *say*, what the media *echoes*, and what capital *does*. We develop a theory of the disagreement between them. Our central result explains *false alarms* endogenously. An institution that trades while addressing a partly credulous crowd optimally talks an asset down while buying it exactly when $\varphi^2 < 2\lambda k < \varphi$, where φ is crowd credulity, λ price impact and k the cost of misreporting. It exaggerates and sells when $\varphi > 1$. A distribution-free identity, $\text{Cov}(r, \text{Say}) = \lambda \text{Cov}(\text{Do}, \text{Say}) - \varphi \sigma_\varepsilon^2$, makes the observable covariance between words and deeds the switch that decides whether words should be followed or faded. Media virality, measured by the Hawkes branching ratio, yields a manipulation dichotomy: echo breeds false alarms in deep markets and exaggeration in shallow ones. When trust is learned adaptively, it converges to truth only if $\lambda k > 1$; a flip bifurcation at $\lambda k = 1$ creates a closed-form credibility cycle, stable for $\frac{5}{6} < \lambda k < 1$, that leads by period-doubling to chaos. For measurement, we prove an exact *semantic declustering* theorem that separates news from echo using timing and embedding similarity. We show that return-aligned contrastive embeddings are affine isometries of market-outcome space, with a Pinsker certificate for imperfect training. We also show that Lévy area identifies whether money moved before the narrative. Monte Carlo simulation confirms every closed form.

1 Introduction

A respected bank publishes a gloomy call. Headlines multiply. The stock falls, though by less than the noise suggests, and weeks later it trades higher. In hindsight, someone was buying all along (Figure 1a). Practitioners call such episodes *false alarms*. Proving intent is neither possible nor necessary. What is possible is to measure the disagreement between three observable voices: **Say** (institutional statements such as ratings, notes and management tone), **Echo** (media repetition and amplification) and **Do** (revealed allocation: holdings, insider trades, options flow and price).

We develop a theory of that disagreement and of how to trade it. Two principles organise the paper. *Deeds over words*: when **Say** and **Do** disagree, the optimal forecast is proportional to their weighted gap, and the *sign* of their covariance decides whether words are followed or faded. *Fade the echo, follow the news*: repetition carries no information, so its price effect must unwind, whereas under-weighted news drifts. The theory never prescribes blind contrarianism; it identifies which cell of Figure 1b applies.

Contributions. The results we believe to be new are:

- **Endogenous false alarms** (Section 4). A closed-form theory of optimal speech while trading, with a four-regime taxonomy: truth, shading, false alarm, and exaggeration with distribution (Theorem 7 and Corollary 8).
- **The Say–Do covariance identity** (Theorem 9), which holds for any distribution and turns an observable statistic into a follow-or-fade switch.
- **A manipulation dichotomy under virality** (Proposition 10) and **credibility cycles** under adaptive trust (Theorem 11), with explicit bifurcation thresholds $\lambda k = 1$ and $\lambda k = \frac{5}{6}$.
- **Semantic declustering** (Theorem 13): exact, factorised news-versus-echo posteriors that combine timing with embedding similarity.
- **Embedding geometry with a certificate** (Theorems 14

and 15), **absorption as an interpretation gap** (Theorem 5), and **lead–lag via Lévy area** (Theorem 16).

The Gaussian follow-or-fade and Say–Do gap results (Theorems 1 and 4) are clean multi-channel statements of Bayesian updating. Regime detection, conformal calibration and Kelly sizing (Appendix B) are classical tools, included so that the paper is self-contained. Proofs are in Appendix A.

2 Related Work

Sentiment and returns. Media pessimism predicts price pressure followed by reversal [43]. Financial text needs domain-specific measurement [35], sentiment matters most where arbitrage is hard [4, 15, 42], and attention drives demand [5, 14]. Prices drift after headlines and reverse after no-news moves [10], heterogeneous agents reconcile under- and overreaction [24], and the same news is read differently across states [45]. Text factors and language models predict returns [1, 27, 34], subject to look-ahead bias [20].

Informed trading and communication. Order flow reveals private information [21, 30]. Volume–return dynamics separate informed from hedging trades [32], predatory traders exploit predictable flow [9], short sellers process news better [16], insider trades split into informative and routine [12], and promotional articles move prices [29]. Cheap talk [13], costly talk with credulous receivers [26], manipulation by informed “gurus” [6] and rumour-based trading [44] study how informed agents shape beliefs.

Mathematical tools. We use Hawkes processes [22] and their cluster representation [23], which underlies stochastic declustering [46], with finance applications in [2]. We also use contrastive learning [39, 41] and the linear representation hypothesis [40]; Lévy area and path signatures [11, 31, 36]; cobweb learning and period-doubling [17, 18, 37]; and the square-root impact law [8].

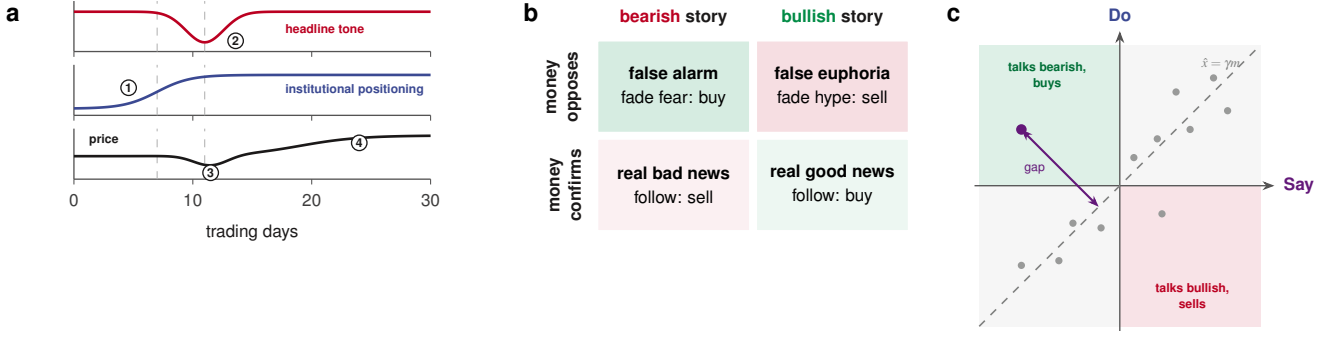


Fig. 1 | Three voices, one decision. **a**, The false-alarm pattern (stylised): positioning builds (1) before the alarm (2); price falls far less than the alarm implies (3), then recovers (4). **b**, The narrative never decides the trade on its own; its agreement with money and price does. **c**, The Say-Do plane. The optimal forecast decreases with signed distance from the line $\hat{x} = \gamma m$ (Theorem 4); the top-left quadrant is the false-alarm zone.

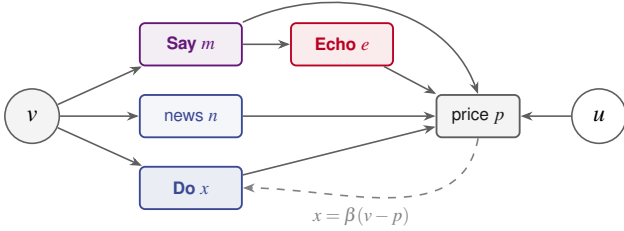


Fig. 2 | Information structure. Echo is a downstream garbling of institutional speech, news is an independent signal of value, and informed demand responds to the price (dashed).

3 A Three-Voice Market

Assumption 1 (Signals). Value is $v \sim \mathcal{N}(0, \sigma_v^2)$, and public channels are $\mathbf{y} = \mathbf{h}v + \mathbf{v} \in \mathbb{R}^K$ with $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \Sigma_v)$ independent of v and $\Sigma_y = \sigma_v^2 \mathbf{h}\mathbf{h}^\top + \Sigma_v \succ 0$.

Assumption 2 (Prices). Noise demand $u \sim \mathcal{N}(0, \sigma_u^2)$ is independent of everything else. Informed demand is $x = \beta(v - p)$ with $\beta > 0$, and the price is $p = \boldsymbol{\phi}^\top \mathbf{y} + \lambda(x + u)$ with $\lambda > 0$, where $\boldsymbol{\phi}$ holds the crowd's narrative multipliers.

Write $\kappa = (1 + \lambda\beta)^{-1}$, $\mathbf{w}^* = \sigma_v^2 \Sigma_y^{-1} \mathbf{h}$ (the Bayes weights, $\mathbb{E}[v | \mathbf{y}] = \mathbf{w}^{*\top} \mathbf{y}$), $\sigma_{v|y}^2 = \text{Var}(v | \mathbf{y})$, and $r = v - p$. Solving the fixed point gives $r = \kappa(v - \boldsymbol{\phi}^\top \mathbf{y} - \lambda u)$ (Figure 2).

3.1 Follow or fade

Theorem 1 (Follow-or-fade). Under Assumptions 1 and 2, $\mathbb{E}[r | \mathbf{y}] = \kappa(\mathbf{w}^* - \boldsymbol{\phi})^\top \mathbf{y}$ and $\text{Var}(r | \mathbf{y}) = \kappa^2(\sigma_{v|y}^2 + \lambda^2 \sigma_u^2) =: \zeta^2$. The loading on channel k is negative (fade) iff $\phi_k > w_k^*$.

Corollary 2 (Fade the echo). If $e = \mathbf{a}^\top \mathbf{y}_{-e} + \xi$ with ξ independent noise of positive variance, then $w_e^* = 0$, and the loading on e is exactly $-\kappa\phi_e$.

Corollary 3 (Follow the news). If $n = v + v_n$ with independent noise v_n , then $w_n^* \in (0, 1)$, and returns drift with n whenever $\phi_n < w_n^*$.

3.2 Deeds over words

Theorem 4 (Say-Do gap optimality). Suppose a positioning proxy $\hat{x} = x + \xi_x$ with $\text{Var} \xi_x = \sigma_x^2$ is also observed. Let $c_x = \beta \zeta^2 / (\beta^2 \zeta^2 + \sigma_x^2)$ and $\theta = \sigma_x^2 / (\beta^2 \zeta^2 + \sigma_x^2)$. Then

$$\mathbb{E}[r | \mathbf{y}, \hat{x}] = \theta \kappa (\mathbf{w}^* - \boldsymbol{\phi})^\top \mathbf{y} + c_x \hat{x}, \quad \text{Var}(r | \mathbf{y}, \hat{x}) = \theta \zeta^2.$$

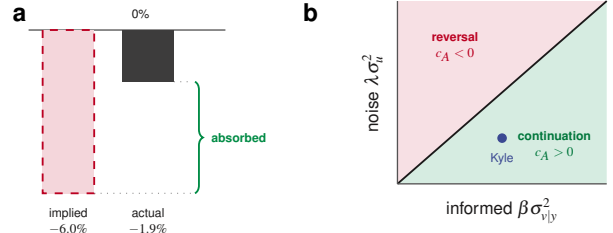


Fig. 3 | Absorption. **a**, The narrative implied a 6% fall; the price fell 1.9%. **b**, Phase diagram of Theorem 5: absorption signals a hidden buyer only where informed intensity dominates noise.

If the Say channel $m = y_1$ is overweighted ($\phi_1 > w_1^*$), the forecast depends on $(m, \hat{x})$ only through $-c_x(\gamma m - \hat{x})$, with $\gamma = \theta \kappa (\phi_1 - w_1^*) / c_x > 0$.

3.3 Absorption: informed or noise?

The absorption of a narrative is the price move net of the narrative-implied move, $a = p - \boldsymbol{\phi}^\top \mathbf{y}$.

Theorem 5 (Absorption sign). $\mathbb{E}[r | \mathbf{y}, a] = \mathbb{E}[r | \mathbf{y}] + c_A(a - \mathbb{E}[a | \mathbf{y}])$ with

$$c_A = \frac{\beta \sigma_{v|y}^2 - \lambda \sigma_u^2}{\lambda (\beta^2 \sigma_{v|y}^2 + \sigma_u^2)}.$$

Absorption predicts continuation iff $\beta \sigma_{v|y}^2 > \lambda \sigma_u^2$, and reversal otherwise. The limits are $c_A = 1/(\lambda\beta)$ as $\sigma_u \rightarrow 0$ and $c_A = -1$ as $\beta \rightarrow 0$.

Lemma 6 (Robustness). For any $\hat{\boldsymbol{\phi}}$, $\hat{a} = p - \hat{\boldsymbol{\phi}}^\top \mathbf{y}$ generates the same σ -field as a given $\mathbf{y}$, so $\mathbb{E}[r | \mathbf{y}, \hat{a}] = \mathbb{E}[r | \mathbf{y}, a]$.

At Kyle's equilibrium intensities [30], $c_A = \frac{1}{2}$. Theorem 5 is in the spirit of Llorente et al. [32], but it conditions on the narrative (Figure 3). In practice, the narrative-implied move comes from the embedding-based interpretation model of Section 5.2, and Lemma 6 protects the signal against misestimating it.

4 Strategic Speech and False Alarms

We now let the institution choose what to say.

Assumption 3 (Strategic institution). An institution knows v , which may have any distribution with mean 0 and variance $\sigma_v^2 < \infty$. It publishes m and trades x . The public sees

$\tilde{m} = m + \varepsilon$, where ε has mean zero and variance σ_ε^2 and is independent of v . The price is $p = \varphi \tilde{m} + \lambda x$, where $\varphi \geq 0$ is the crowd's credulity. Misreporting costs $\frac{k}{2}(m - v)^2$. The institution maximises $J = x(v - \varphi m - \lambda x) - \frac{k}{2}(m - v)^2$. Let $a := 2\lambda k$.

Assumption 4 (Echo amplification). Each article moves the price by φ_0 per unit of message, and a Hawkes cascade with branching ratio $n \in [0, 1)$ repeats it. Hence $\varphi = \varphi_0/(1 - n)$ (Proposition 12).

Theorem 7 (Optimal speech and trade). J has a unique maximiser iff $a > \varphi^2$, namely $m^* = \psi v$ and $x^* = \chi v$ with

$$\psi = \frac{a - \varphi}{a - \varphi^2}, \quad \chi = \frac{k(1 - \varphi)}{a - \varphi^2}.$$

If $a < \varphi^2$, J is unbounded: only position limits contain manipulation.

Corollary 8 (Taxonomy). Let $a > \varphi^2$ and $v > 0$. The institution tells the **truth** if $\varphi = 1$. It **shades** ($0 < \psi < 1$, buys) if $\varphi < 1 < a/\varphi$. It raises a **false alarm** ($\psi < 0 < \chi$: talks down, buys) if $\varphi^2 < a < \varphi < 1$. It **exaggerates and sells** ($\psi > 1$, $\chi < 0$) if $\varphi > 1$.

Theorem 9 (Say–Do covariance identity). Under Assumption 3, let the message rule $m(v)$ be arbitrary and the trade optimal given it. Then, pathwise, $r = \lambda x - \varphi \varepsilon$. Hence, for any law of v ,

$$\text{Cov}(r, \tilde{m}) = \lambda \text{Cov}(x, \tilde{m}) - \varphi \sigma_\varepsilon^2, \quad \mathbb{E}[r | x] = \lambda x.$$

In equilibrium, $\text{Cov}(x, \tilde{m}) = \psi \chi \sigma_v^2$, which is negative exactly in the false-alarm and exaggeration regimes.

Why it matters. Both sides of the identity are estimable, and no distributional assumption is needed. The rolling sign of $\widehat{\text{Cov}}(\text{Say}, \text{Do})$ tells the econometrician whether to follow or fade institutional words, while deeds are always followed, with slope λ .

Proposition 10 (Manipulation dichotomy). Under Assumptions 3 and 4 with $\varphi_0 \in (0, 1)$, call a regime manipulative if $\text{Cov}(\text{Say}, \text{Do}) < 0$. (i) If $a < 1$ (a deep market), the only manipulative regime is the false alarm, which occurs iff $\max\{0, 1 - \varphi_0/a\} < n < 1 - \varphi_0/\sqrt{a}$. (ii) If $a > 1$ (a shallow market), it is exaggeration, which occurs iff $1 - \varphi_0 < n < 1 - \varphi_0/\sqrt{a}$. (iii) For $n > 1 - \varphi_0/\sqrt{a}$ no interior optimum exists.

Adaptive trust. Suppose the crowd resets its credulity each round to the least-squares slope of value on message under the previous strategy, a cobweb rule [17, 18]. This gives $\varphi_{t+1} = 1/\psi(\varphi_t) = F_a(\varphi_t)$ with $F_a(\varphi) = (a - \varphi^2)/(a - \varphi)$.

Theorem 11 (Credibility cycles). (i) For $a > 1$, truth ($\varphi = 1$) is the unique admissible fixed point. (ii) $F'_a(1) = -1/(a - 1)$, so truth is stable iff $\lambda k > 1$. (iii) For $1 < a < 2$ there is exactly one 2-cycle,

$$\varphi_\pm = \frac{1}{2}(a \pm \sqrt{a(2 - a)}), \quad \varphi_- < 1 < \varphi_+,$$

with multiplier $(5a - 9)/(a - 1)$. It is stable iff $\frac{5}{6} < \lambda k < 1$, and it alternates between shading and exaggeration. (iv) The cycle is born by a flip at $a = 2$ and loses stability by a second flip at $a = \frac{5}{3}$.

Relation to prior work. Bénabou and Laroque [6] derive credibility dynamics from Bayesian learning about a random honest type, and Kartik et al. [26] study costly talk with credulous receivers. Our contribution is a trading-coupled linear–quadratic model in which the false-alarm region, the Say–Do covariance identity and the bifurcation thresholds are all explicit and estimable.

5 Measuring the Voices

5.1 Echo: exact semantic declustering

Articles arrive at times $\tau_1 < \dots < \tau_N$ with embeddings $\mathbf{z}_j$ and a Hawkes intensity $\lambda(t) = \mu + \sum_{\tau_i < t} g(t - \tau_i)$ with $n = \int g < 1$. In the cluster representation [23], each article is either news (an immigrant) or an echo of an earlier article.

Proposition 12 (Echo share). In stationarity, the long-run share of articles that are echoes equals the branching ratio n , and the mean cascade size is $(1 - n)^{-1}$.

Assumption 5 (Marked cascade). News embeddings are i.i.d. with density f_0 . An echo of article l has density $f(\cdot | \mathbf{z}_l)$, for example von Mises–Fisher, $f \propto e^{\kappa \langle \mathbf{z}, \mathbf{z}_l \rangle}$.

Theorem 13 (Semantic declustering). Given all times and embeddings on $[0, T]$, the parent labels $\pi(j)$ are independent, with

$$\mathbb{P}(\pi(j) = 0 | \cdot) = \frac{\mu f_0(\mathbf{z}_j)}{\Lambda_j},$$

$$\mathbb{P}(\pi(j) = l | \cdot) = \frac{g(\tau_j - \tau_l) f(\mathbf{z}_j | \mathbf{z}_l)}{\Lambda_j},$$

where $\Lambda_j = \mu f_0(\mathbf{z}_j) + \sum_{l < j} g(\tau_j - \tau_l) f(\mathbf{z}_j | \mathbf{z}_l)$. The MMSE split of sentiment $s_j = \langle \mathbf{w}_s, \mathbf{z}_j \rangle$ is $s_j^{\text{new}} = \sum_j p_j^{\text{sem}} s_j$ and $s_j^{\text{echo}} = \sum_j (1 - p_j^{\text{sem}}) s_j$, with $p_j^{\text{sem}} = \mu f_0(\mathbf{z}_j)/\Lambda_j$. Semantics never increases the expected posterior entropy of parentage relative to timing alone.

Setting $f \equiv f_0$ recovers timing-only stochastic declustering [46], where $p_j = \mu/\lambda(\tau_j)$. Combined with Corollary 2, s_j^{echo} estimates exactly the garbled component that the crowd should not have priced (Figure 5).

5.2 Market-aligned embeddings

Documents are embedded on the sphere, $\mathbf{z}_j \in S^{D-1}$. Each has a post-event response vector $\boldsymbol{\rho}_j$ (multi-horizon abnormal returns, abnormal volume, change in implied volatility). Let $S_{jk} = \langle \mathbf{z}_j, \mathbf{z}_k \rangle$ and $D_{jk} = \|\boldsymbol{\rho}_j - \boldsymbol{\rho}_k\|^2$. The return-aligned soft-contrastive loss is $\mathcal{L}(Z) = -\sum_j \sum_{k \neq j} q_{jk} \log p_{jk}(Z)$, where

$$q_{jk} \propto e^{-D_{jk}/2b^2}, \quad p_{jk}(Z) \propto e^{S_{jk}/\tau},$$

and both are normalised over $k \neq j$.

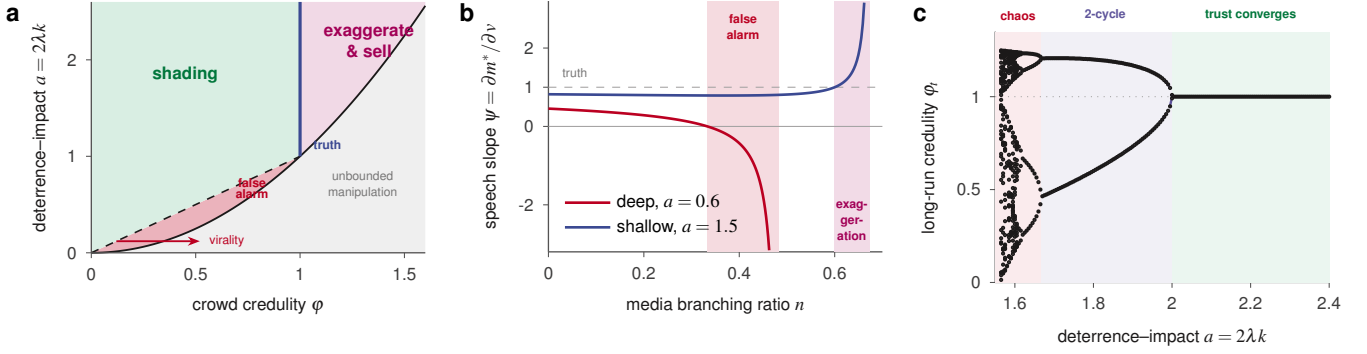


Fig. 4 | The economics of false alarms. **a**, Regime map (Theorem 7 and Corollary 8). False alarms occupy the wedge $\phi^2 < a < \phi$; echo amplification moves the market to the right. **b**, Speech slope against the media echo share (Proposition 10, $\phi_0 = 0.4$). Deep markets pass through a false-alarm window ($\psi < 0$); shallow markets pass through an exaggeration window ($\psi > 1$). **c**, Bifurcation diagram of adaptive trust (Theorem 11). Dots show the long-run attractor, and dashed curves the closed-form cycle $\phi_{\pm}$. Trust converges for $a > 2$, cycles for $\frac{5}{3} < a < 2$, and becomes chaotic below (numerical Lyapunov exponent ≈ 0.10 at $a = 1.6$).

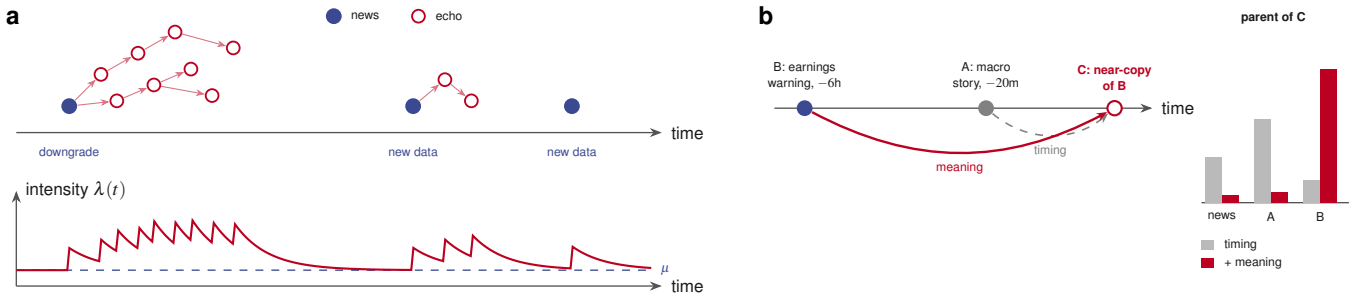


Fig. 5 | Separating news from echo. **a**, A news article (blue) triggers a cascade of echoes (red), and the intensity jumps and decays to the baseline μ . The long-run echo share equals the branching ratio (Proposition 12). **b**, Semantic declustering (Theorem 13; illustrative numbers). Timing alone attributes article C to the most recent story A; adding embedding similarity correctly identifies C as an echo of B.

Theorem 14 (Minimisers are isometries of outcome space). $\mathcal{L}(Z) \geq \sum_j H(q_j)$, with equality iff $S_{jk} = c - \frac{\tau}{2b^2} D_{jk}$ for some constant c and all $j \neq k$. At equality, $\|\mathbf{z}_j - \mathbf{z}_k\|^2 = 2(1 - c) + \frac{\tau}{b^2} \|\boldsymbol{\rho}_j - \boldsymbol{\rho}_k\|^2$, so embedding neighbours are exactly outcome neighbours.

Theorem 15 (Robust neighbour preservation). For any Z , let $\varepsilon_j = \text{KL}(q_j \| p_j(Z))$. If $q_{jk} - q_{jl} > \sqrt{2\varepsilon_j}$, then $\|\mathbf{z}_j - \mathbf{z}_k\| < \|\mathbf{z}_j - \mathbf{z}_l\|$.

Theorem 15 turns the leftover training loss into a certificate of which analog rankings are reliable. This justifies “d  j   vu” forecasting of the narrative-implied reaction by averaging outcomes of the nearest historical analogues (Appendix B). Sentiment is read as a projection onto a topic-orthogonal direction $\mathbf{w}_s^\perp = (I - UU^\top)\mathbf{w}_s$, which is invariant to shifts along topic directions U . Surprise is measured against an expected embedding $\hat{\mathbf{z}}$ as a Mahalanobis distance, which is χ_D^2 -calibrated (Figure 6).

5.3 Who moved first? L  vy area

For paths X, Y , the L  vy area $\mathcal{A}_T = \frac{1}{2} \int_0^T (X_u - X_0) dY_u - (Y_u - Y_0) dX_u$ is the antisymmetric part of the level-two signature. For a closed loop it is the signed enclosed area.

Theorem 16 (Expected area identifies lead-lag). Let (X, Y) be zero-mean, jointly stationary and mean-square C^1 , with

cross-covariance $C(h) = \mathbb{E}[X_s Y_{s+h}] \in C^1$. Then

$$\mathbb{E}[\mathcal{A}_T] = TC'(0) - \frac{1}{2}(C(T) - C(-T)),$$

so $\mathbb{E}[\mathcal{A}_T]/T \rightarrow C'(0)$ when C is bounded. If $Y_t = \gamma X_{t-\ell} + Z_t$, with Z independent and X having a unimodal autocovariance, then the limiting rate has the sign of $\gamma\ell$.

With X = positioning and Y = headline gloom, a positive area rate means that money moved first and the story turned against it (Figure 7).

6 From Signals to Decisions

Events belong to latent regimes (informational, overreaction, strategic) with emissions $\mathbf{o}_t = (s^{\text{new}}, s^{\text{echo}}, G, A, \mathcal{A}, \dots)$. Filtered regime probabilities are causal, and the MSE-optimal forecast is the mixture of regime experts. Strategic episodes of n events are detected with error at most $\frac{1}{2}e^{-n\Delta^2/8}$. Adaptive conformal intervals attain long-run coverage deterministically [19]. Positions use the shrunk Kelly fraction $c^* = \text{SNR}/(1 + \text{SNR})$, since full Kelly destroys growth when the edge is noisier than it is large (Appendix B). A transparent composite whose every sign is fixed by a theorem is

$$D = \omega_1 s^{\text{new}} - \omega_2 s^{\text{echo}} + \omega_3 \text{Do} + \widehat{\text{sgn}} \omega_4 \gamma \text{Say} + \omega_5 A + \omega_6 \mathcal{A}^\pm.$$

Here $\widehat{\text{sgn}}$ is the rolling sign of $\text{Cov}(\text{Say}, \text{Do})$ (Theorem 9): deeds are always followed, and words are followed when the

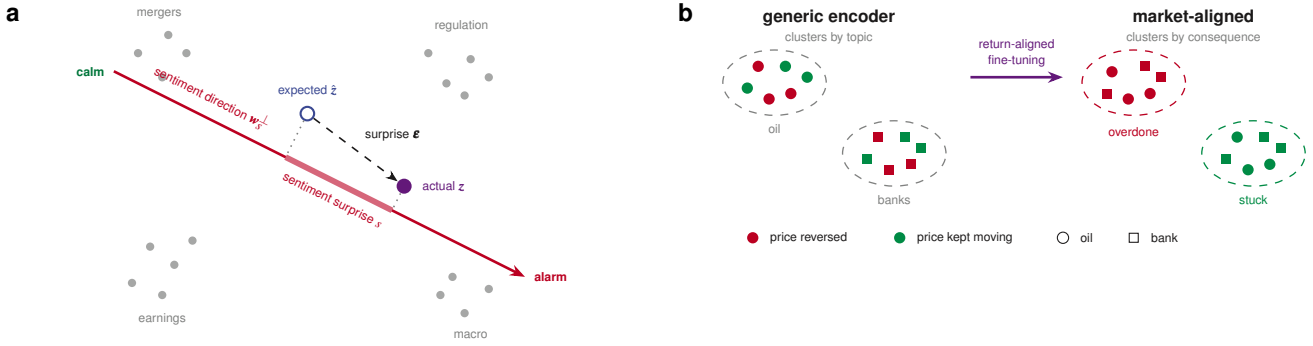


Fig. 6 | A map of meaning aligned with markets. **a**, Embeddings cluster by topic, sentiment is a direction, and surprise is measured against the *expected* story; its projection onto the alarm direction is the sentiment surprise. **b**, At the optimum of the return-aligned objective (Theorem 14), distances mirror market aftermaths, so “news that got overdone” becomes a region of the map.

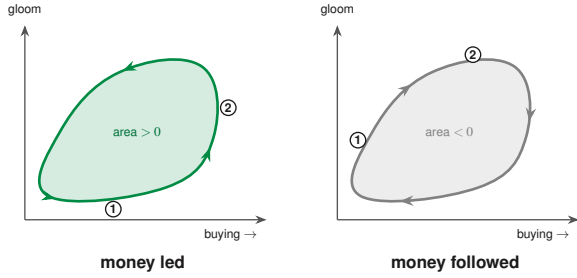


Fig. 7 | Lead-lag as a loop. Traced in the (positioning, gloom) plane, the orientation of the path shows who moved first, and its enclosed area shows how strongly (Theorem 16).

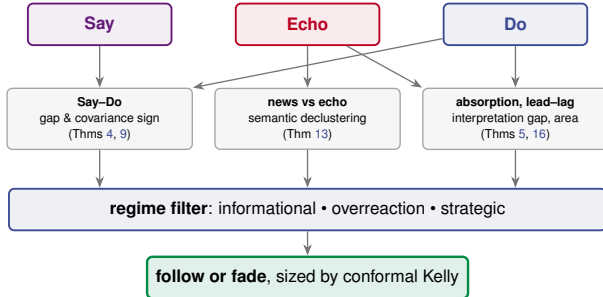


Fig. 8 | Architecture. Three voices produce three families of theorem-signed signals, and a regime filter turns them into a single calibrated decision.

covariance is positive and faded when it is negative, in which case the two terms combine into $-G_\gamma$ (Theorem 4). $\mathcal{A}^\pm$ is the Lévy area signed by the direction of positioning. Figure 8 shows the complete architecture.

7 Numerical Verification

Every closed form was checked independently (Table 1). Gaussian results were checked by regression on 2×10^6 simulated draws. Theorem 9 was checked with fat-tailed values (t_3) and Laplace noise, which tests its distribution-free claim. The optimal speech of Theorem 7 was checked against numerical optimisation. Theorem 13 was checked by brute-force enumeration of all parent labellings, and Theorem 16 on a smooth Gaussian process with a known lag.

Table 1 | Theory versus simulation. Theory values are exact; simulations use 2×10^6 draws (seed 0).

Result	Quantity	Theory	Sim.
Corollary 2	Bayes weight on echo	0	-0.0006
Theorem 1	loading, echo channel	-0.1840	-0.1843
Theorem 4	loading on $\hat{x}$ (c_x)	0.4012	0.4004
Theorem 5	absorption coefficient c_A	-0.1997	-0.2004
Theorem 7	(m^*, x^*) , $\varphi=0.6$, $a=0.5$	(-0.714, 1.429)	(-0.714, 1.429)
Theorem 9	$\text{Cov}(r, \tilde{m})$, t_3 values	-2.294	-2.293
Theorem 9	idem, exaggeration	-7.608	-7.676
Theorem 11	2-cycle at $a=1.95$	{0.819, 1.131}	{0.819, 1.131}
Theorem 11	multiplier, $a=1.8$	0	superstable
Theorem 13	max posterior error	0	1.1×10^{-16}
Theorem 16	area rate, lag $\ell = 0.5$	0.2356	0.2360
Theorem 21	growth at $c^* = 0.338$	0.01056	0.01055
Theorem 21	growth at full Kelly	-0.0300	-0.0300

Table 2 | Illustration: a hypothetical downgrade (numbers are illustrative, not data).

Signal (result)	Reading
news / echo sentiment (Thm 13)	-0.5σ / -2.3σ
echo share n (Prop. 12)	0.82
Say-Do covariance (Thm 9)	negative: fade Say
implied vs actual move (Thm 5)	-6.0% vs -1.9%
Lévy area rate (Thm 16)	positive: money led
regime I / O / S	0.10 / 0.28 / 0.62
Kelly fraction (Thm 21), $\text{SNR} \approx 0.5$	$c^* \approx 0.33$

Reading: probable false alarm; lean long at one third of full Kelly.

8 Discussion

Testable predictions. The theory predicts:

- opposite signs on s^{new} and s^{echo} ;
- a negative Say-Do covariance in episodes later identified as manipulative;
- false alarms concentrated at intermediate echo shares in liquid markets, and exaggeration in illiquid ones;
- negative autocorrelation of the rolling Say-Do covariance where λk is moderate;
- an absorption loading whose sign flips with flow toxicity;
- higher returns after positive Lévy area.

The cleanest first test pairs management tone on earnings calls (**Say**) with insider trades (**Do**), both of which are public.

Limitations. Crowd pricing is reduced-form, and credulity is exogenous within a round. The price-formation results fix signs and sufficient statistics, not magnitudes. **Theorem 16** assumes smooth, stationary paths. Empirical validation with point-in-time data and purged cross-validation [3, 33] is left to companion work.

Ethics. A strategic reading means that narrative and capital moved in opposite directions in a particular order. It is not evidence of deception, and firms maintain information barriers between research and trading. The framework uses public information and lawful disclosures only. It must never be used to create or spread narratives intended to move prices.

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A Proofs

Lemma 17 (Gaussian conditioning). *If $(\mathbf{a}, \mathbf{b})$ is jointly Gaussian with $\text{Var} \mathbf{b} \succ 0$, then $\mathbb{E}[\mathbf{a} | \mathbf{b}] = \mathbb{E} \mathbf{a} + \text{Cov}(\mathbf{a}, \mathbf{b}) \text{Var}(\mathbf{b})^{-1}(\mathbf{b} - \mathbb{E} \mathbf{b})$ and $\text{Var}(\mathbf{a} | \mathbf{b}) = \text{Var} \mathbf{a} - \text{Cov}(\mathbf{a}, \mathbf{b}) \text{Var}(\mathbf{b})^{-1} \text{Cov}(\mathbf{b}, \mathbf{a})$. The same holds conditionally on a further jointly Gaussian vector.*

Proof. The residual $\mathbf{e} = \mathbf{a} - \mathbb{E} \mathbf{a} - \text{Cov}(\mathbf{a}, \mathbf{b}) \text{Var}(\mathbf{b})^{-1}(\mathbf{b} - \mathbb{E} \mathbf{b})$ is uncorrelated with $\mathbf{b}$, and therefore independent of it by joint Gaussianity. Its variance expands to the stated formula. $\square$

Price and return. From $x = \beta(v - p)$, $p(1 + \lambda\beta) = \boldsymbol{\varphi}^\top \mathbf{y} + \lambda\beta v + \lambda u$, so $r = v - p = \kappa(v - \boldsymbol{\varphi}^\top \mathbf{y} - \lambda u)$.

Proof of Theorem 1. $\mathbb{E}[v | \mathbf{y}] = \sigma_v^2 \mathbf{h}^\top \Sigma_y^{-1} \mathbf{y} = \mathbf{w}^{\star\top} \mathbf{y}$ and $\mathbb{E}[u | \mathbf{y}] = 0$. The residual $r - \mathbb{E}[r | \mathbf{y}] = \kappa((v - \mathbb{E}[v | \mathbf{y}]) - \lambda u)$ has independent components given $\mathbf{y}$. $\square$

Proof of Corollary 2. $\sigma(\mathbf{y}) = \sigma(\mathbf{y}_{-e}, \xi)$ with ξ independent of $(v, \mathbf{y}_{-e})$, so $\mathbb{E}[v | \mathbf{y}] = \mathbb{E}[v | \mathbf{y}_{-e}]$, which is linear in $\mathbf{y}_{-e}$ alone. If $\mathbf{w}^{\star\top} \mathbf{y} = \mathbf{w}'^\top \mathbf{y}$ almost surely with $\mathbf{w}'_e = 0$, then $(\mathbf{w}^* - \mathbf{w}')^\top \Sigma_y (\mathbf{w}^* - \mathbf{w}') = 0$, and $\Sigma_y \succ 0$ gives $\mathbf{w}^* = \mathbf{w}'$. $\square$

Proof of Corollary 3. Conditionally on $\mathbf{y}_{-n}$, $v \sim \mathcal{N}(\mu', s^2)$ with $s^2 > 0$. The scalar update puts weight $s^2/(s^2 + \sigma_n^2) \in (0, 1)$ on n . Apply Theorem 1. $\square$

Proof of Theorem 4. $x = \beta r$, so $\hat{x} = \beta r + \xi_x$. Given $\mathbf{y}$, $\text{Cov}(r, \hat{x}) = \beta \zeta^2$ and $\text{Var} \hat{x} = \beta^2 \zeta^2 + \sigma_x^2$. Lemma 17 gives $\mathbb{E}[r | \mathbf{y}, \hat{x}] = (1 - \beta c_x) \mathbb{E}[r | \mathbf{y}] + c_x \hat{x}$ with $1 - \beta c_x = \theta$, and $\text{Var} = \zeta^2 - c_x \beta \zeta^2 = \theta \zeta^2$. The $(m, \hat{x})$ terms are $-\theta \kappa(\boldsymbol{\varphi}_1 - \mathbf{w}_1^*) m + c_x \hat{x} = -c_x (\gamma m - \hat{x})$. $\square$

Proof of Theorem 5. Let $\tilde{v} = v - \mathbb{E}[v | \mathbf{y}]$. Then $r - \mathbb{E}[r | \mathbf{y}] = \kappa(\tilde{v} - \lambda u)$. Since $a = \lambda \beta r + \lambda u$, we get $a - \mathbb{E}[a | \mathbf{y}] = \lambda \beta \kappa \tilde{v} + \lambda(1 - \lambda \beta \kappa)u = \lambda \kappa(\beta \tilde{v} + u)$, using $1 - \lambda \beta \kappa = \kappa$. Hence $\text{Cov}(r, a | \mathbf{y}) = \lambda \kappa^2(\beta \sigma_{v|y}^2 - \lambda \sigma_u^2)$ and $\text{Var}(a | \mathbf{y}) = \lambda^2 \kappa^2(\beta^2 \sigma_{v|y}^2 + \sigma_u^2)$. Their ratio is c_A by Lemma 17. At Kyle's intensities, $\beta \sigma_{v|y}^2 = \sigma_u \sigma_{v|y}$ and $\lambda \sigma_u^2 = \frac{1}{2} \sigma_u \sigma_{v|y}$, which give $c_A = \frac{1}{2}$. $\square$

Proof of Lemma 6. $\hat{a} = a + (\boldsymbol{\varphi} - \hat{\boldsymbol{\varphi}})^\top \mathbf{y}$ is an invertible affine map of a given $\mathbf{y}$. $\square$

Proof of Theorem 7. J is quadratic with Hessian $H = \begin{pmatrix} -2\lambda & -\varphi \\ -\varphi & -k \end{pmatrix}$, which is negative definite iff $\det H = a - \varphi^2 > 0$. Then J is strictly concave, and its unique maximiser solves $v - \varphi m - 2\lambda x = 0$ and $-\varphi x - k(m - v) = 0$. Substitution gives $m(a - \varphi^2) = v(a - \varphi)$, and $1 - \varphi \psi = a(1 - \varphi)/(a - \varphi^2)$ gives χ . If $\det H < 0$, then $J(\mathbf{td}) \rightarrow \infty$ along an eigenvector $\mathbf{d}$ with positive eigenvalue. $\square$

Proof of Corollary 8. $\psi > 0 \iff a > \varphi$; $\psi < 1 \iff \varphi < 1$; and $\text{sgn } \chi = \text{sgn}(1 - \varphi)$. If $\varphi > 1$, then $a > \varphi^2 > \varphi$, so $\psi > 1$. $\square$

Proof of Theorem 9. Given m , the optimal trade is $x = (v - \varphi m)/(2\lambda)$, so $v - \varphi m = 2\lambda x$ and $r = v - \varphi(m + \varepsilon) - \lambda x = \lambda x - \varphi \varepsilon$. Since x and m are functions of v , which is independent of ε , we have $\text{Cov}(\varepsilon, \tilde{m}) = \sigma_\varepsilon^2$ and $\mathbb{E}[\varepsilon | x] = 0$. In equilibrium, $\text{Cov}(\chi v, \psi v) = \psi \chi \sigma_v^2$, and $\psi \chi < 0$ exactly in the false-alarm and exaggeration regimes. $\square$

Proof of Proposition 10. $\varphi = \varphi_0/(1 - n)$ increases from φ_0 to ∞ . A false alarm needs $\varphi^2 < a < \varphi < 1$, hence $a < 1$. Exaggeration needs $1 < \varphi < \sqrt{a}$, hence $a > 1$. The conditions $\varphi > a$, $\varphi > 1$ and $\varphi < \sqrt{a}$ translate to $n > 1 - \varphi_0/a$, $n > 1 - \varphi_0$ and $n < 1 - \varphi_0/\sqrt{a}$. For $a < 1$, $\varphi < \sqrt{a}$ already implies $\varphi < 1$. $\square$

Proof of Theorem 11. (i) $F_a(\varphi) = \varphi \iff \varphi = 1$, and admissibility ($\varphi^2 < a$) needs $a > 1$. (ii) $F'_a(\varphi) = (a - 2a\varphi + \varphi^2)/(a - \varphi)^2$, so $F'_a(1) = -1/(a - 1)$, whose modulus is below 1 iff $a > 2$. (iii) Using $a - F_a(\varphi) = (a^2 - a\varphi - a + \varphi^2)/(a - \varphi)$, direct computation gives

$$F_a(F_a(\varphi)) - \varphi = \frac{-a(\varphi - 1)(2\varphi^2 - 2a\varphi + a^2 - a)}{(a - \varphi)^2(a - F_a(\varphi))}.$$

The quadratic factor has roots $\varphi_\pm$, which are real and different from 1 for $1 < a < 2$; its value at $\varphi = 1$ is $(a - 1)(a - 2) \neq 0$. They are admissible: $\varphi_\pm^2 = \frac{a}{2}(1 \pm \sqrt{a(2 - a)}) < a$ because $a(2 - a) < 1$; $\varphi_\pm < a$ because $\sqrt{a(2 - a)} < a \iff a > 1$; and $\psi(\varphi_\pm) > 0$. On the cycle, $\varphi^2 = a\varphi - (a^2 - a)/2$, so the numerator of F'_a equals $a(3 - a - 2\varphi)/2$. With $\varphi_+ + \varphi_- = a$ and $\varphi_+ \varphi_- = (a^2 - a)/2$, we get $(3 - a - 2\varphi_+)(3 - a - 2\varphi_-) = (5a - 9)(a - 1)$ and $(a - \varphi_+)(a - \varphi_-) = a(a - 1)/2$. Hence the multiplier is $(5a - 9)/(a - 1)$, whose modulus is below 1 iff $\frac{5}{3} < a < 2$ when $1 < a < 2$. (iv) At $a = 2$ the multiplier is 1 and $\varphi_\pm = 1$; at $a = \frac{5}{3}$ it is -1 . $\square$

Proof of Proposition 12. Each news article finds a Galton–Watson cluster of expected size $\sum_k n^k = (1 - n)^{-1}$. News arrives at rate μ , so articles arrive at rate $\mu/(1 - n)$, and the news share is $1 - n$. $\square$

Proof of Theorem 13. In the cluster representation, a labelled configuration is a realisation of independent Poisson streams: news at rate $\rho_0 = \mu$, and echoes of l at rate $\rho_l(t) = g(t - \tau_l) \mathbf{1}\{t > \tau_l\}$. Its density is the product of the stream likelihoods and the mark densities,

$$e^{-\int_0^T \lambda} \prod_j \rho_{\pi(j)}(\tau_j) f_{\pi(j)}(\mathbf{z}_j), \quad f_0 \text{ (news)}, \quad f_l = f(\cdot | \mathbf{z}_l),$$

because $\sum_k \rho_k = \lambda$. This factorises over j . Summing over labellings gives $e^{-\int \lambda} \prod_j \Lambda_j$, so the posterior is a product of the stated categorical laws. Because s_j is a function of $\mathbf{z}_j$, $\mathbb{E}[\mathbf{1}\{\pi(j) = 0\} s_j | \cdot] = s_j p_j^{\text{sem}}$. Integrating out the marks recovers the timing-only posterior, and conditioning does not increase expected entropy. $\square$

Proof of Theorem 14. Per anchor, $-\sum_k q_{jk} \log p_{jk} = H(q_j) + \text{KL}(q_j \| p_j)$, with equality to $H(q_j)$ iff $p_j = q_j$, that is, iff the logits differ by a constant c_j : $S_{jk}/\tau + D_{jk}/2b^2 = c_j/\tau$. Symmetry of S and D forces $c_j = c_k$ for every pair, and on the sphere $\|\mathbf{z}_j - \mathbf{z}_k\|^2 = 2 - 2S_{jk}$. $\square$

Proof of Theorem 15. Pinsker gives $\sum_k |p_{jk} - q_{jk}| \leq \sqrt{2\epsilon_j}$, so $p_{jk} - p_{jl} \geq q_{jk} - q_{jl} - \sqrt{2\epsilon_j} > 0$. Softmax is monotone in $S_{j\cdot}$, so $S_{jk} > S_{jl}$, which gives the distance ordering. $\square$

Proof of Theorem 16. Mean-square differentiability gives $\mathbb{E}[X_s \dot{Y}_t] = C'(t - s)$ and $\mathbb{E}[Y_s \dot{X}_t] = -C'(s - t)$. Hence $\mathbb{E}[(X_u - X_0) \dot{Y}_u] = C'(0) - C'(u)$ and $\mathbb{E}[(Y_u - Y_0) \dot{X}_u] = -C'(0) + C'(-u)$. By Fubini,

$$\mathbb{E}[\mathcal{A}_T] = \frac{1}{2} \int_0^T (2C'(0) - C'(u) - C'(-u)) du = TC'(0) - \frac{1}{2} (C(T) - C(-T)).$$

For $Y_t = \gamma X_{t-\ell} + Z_t$, $C(h) = \gamma R(h - \ell)$, so $C'(0) = -\gamma R'(\ell)$. This has the sign of $\gamma\ell$ when R is even with $R' < 0$ on $(0, \infty)$. $\square$

B Additional Results

Proposition 18 (Causal filtering and mixture optimality). *Filtered regime probabilities $\pi_t \propto f(\mathbf{o}_t) \odot P^\top \pi_{t-1}$ are $\mathcal{F}_t$ -measurable, and $\mathbb{E}[r_{t+h} | \mathcal{F}_t] = \sum_k \pi_t(k) \mathbb{E}[r_{t+h} | \mathcal{F}_t, K_t = k]$. Smoothed probabilities are not $\mathcal{F}_t$ -measurable in general and must not be used for forecasting.*

Proof. The recursion is Bayes' rule, and the mixture follows from the tower property. $\square$

Theorem 19 (Exponential detectability). *For n events that are i.i.d. within a regime, with emissions $\mathcal{N}(\mathbf{m}_i, \Sigma)$ and priors π_i , the MAP error satisfies $P_e \leq \sqrt{\pi_0 \pi_1} e^{-n\Delta^2/8} \leq \frac{1}{2} e^{-n\Delta^2/8}$, where $\Delta^2 = (\mathbf{m}_1 - \mathbf{m}_0)^\top \Sigma^{-1} (\mathbf{m}_1 - \mathbf{m}_0)$.*

Proof. $\min\{a, b\} \leq \sqrt{ab}$ gives $P_e \leq \sqrt{\pi_0 \pi_1} \text{BC}^n$ [7, 25]. For equal-covariance Gaussians, completing the square gives $\text{BC} = \int \sqrt{p_0 p_1} = e^{-\Delta^2/8}$. $\square$

Theorem 20 (Adaptive conformal coverage; [19, Prop. 4.1]). *With nested sets $\hat{C}_t(\alpha')$ that equal $\mathbb{R}$ for $\alpha' \leq 0$ and $\emptyset$ for $\alpha' \geq 1$, and the update $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, every outcome sequence satisfies $|\frac{1}{T} \sum_t \text{err}_t - \alpha| \leq (\max\{\alpha_1, 1 - \alpha_1\} + \gamma)/(\gamma T)$.*

Proof (reproduced). By induction, $\alpha_t \in [-\gamma, 1 + \gamma]$. Telescoping gives $\alpha_{T+1} - \alpha_1 = \gamma \sum_t (\alpha - \text{err}_t)$. $\square$

Theorem 21 (Kelly shrinkage under estimation risk; classical). *With excess growth rate $g(f) = f\mu - \frac{1}{2}f^2\sigma^2$ [28, 38], an unbiased estimate $\hat{\mu}$ with variance s^2 that is independent of future returns, and $f = c\hat{\mu}/\sigma^2$, the expected growth is $\mathbb{E}g = (c\mu^2 - \frac{1}{2}c^2(\mu^2 + s^2))/\sigma^2$. It is maximised at $c^* = \mu^2/(\mu^2 + s^2)$. Full Kelly yields $(\mu^2 - s^2)/(2\sigma^2)$, which is negative when $s > |\mu|$.*

Proof. Expand $\mathbb{E}[f\mu]$ and $\mathbb{E}[f^2]$ and maximise the resulting concave quadratic in c . $\square$

Proposition 22 (Analog forecasting). *If $\text{AR}_j = \mu(\mathbf{z}_j) + e_j$ with uncorrelated mean-zero errors of variance at most σ^2 , μ is L -Lipschitz, and weights $\omega_j \geq 0$ sum to one, then the estimator $\hat{\mu}(\mathbf{z}) = \sum_j \omega_j \text{AR}_j$ has bias at most $L \sum_j \omega_j d(\mathbf{z}, \mathbf{z}_j)$ and variance at most $\sigma^2 \sum_j \omega_j^2$.*

Proposition 23 (Calibrated surprise). *If $\mathbf{z} | \mathcal{F} \sim \mathcal{N}(\hat{\mathbf{z}}, \Sigma)$, then $S = (\mathbf{z} - \hat{\mathbf{z}})^\top \Sigma^{-1} (\mathbf{z} - \hat{\mathbf{z}}) \sim \chi_D^2$, and S equals Shannon surprisal up to additive constants.*

C Implementation Notes

Encoder. A transformer sentence encoder [41] with low-rank adapters, trained on the return-aligned loss plus a KL anchor to the frozen encoder, on rolling windows whose outcome horizons close before each refit date. Pretrained components must predate each test window [20].

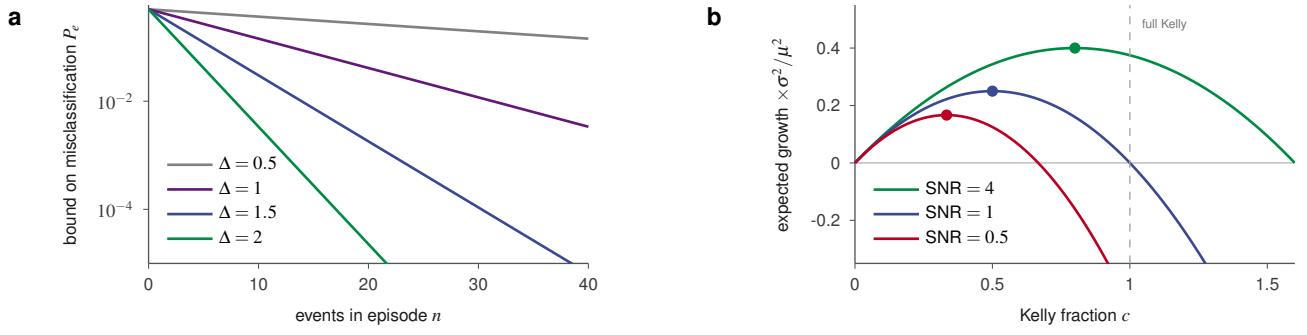


Fig. 9 | Detection and sizing. **a.** Misclassification of strategic episodes decays exponentially in episode length (Theorem 19). **b.** Expected growth as a function of the Kelly fraction; dots mark $c^* = \text{SNR}/(1 + \text{SNR})$ (Theorem 21).

Echo. Exponential Hawkes kernels are fitted per semantic cluster, with von Mises–Fisher echo marks. The posteriors of Theorem 13 cost $O(N \cdot \bar{k})$ with $\bar{k}$ candidate parents retrieved from a vector index.

Say–Do. Pair management tone with insider trades, and published house views with disclosed holdings or futures positioning. A lag-augmented Kalman filter merges disclosures of mixed frequency, and a rolling window estimates $\text{Cov}(\text{Say}, \text{Do})$.

Validation. Point-in-time data, combinatorial purged cross-validation and deflated performance statistics [3, 33].